

MSc 511 --- AOL ASSIGNMENT

- 1) Given the equilibrium on the sheet, what are the price elasticity of supply and the price elasticity of demand at the current price (This is a numerical ± 1 in the price, multiplied by the appropriate ratio)? From a consumer standpoint what kind of good is it?

Ans.

The current price is \$ 6.2166. By changing the price $\pm .1$, the ΔW is 2,046.9311.

$$W = \left(\frac{\alpha + \beta}{K} P \right)^{\frac{\alpha + \beta}{1 - \alpha - \beta}}$$

$$\frac{\partial W}{\partial P} = \left(\frac{\alpha + \beta}{1 - \alpha - \beta} \right) * \left(\frac{\alpha + \beta}{K} * P \right)^{\frac{\alpha + \beta - (1 - \alpha - \beta)}{1 - \alpha - \beta}} * \frac{\alpha + \beta}{K} = 10210.38604$$

$$\text{The price elasticity of supply: } \varepsilon_s = \left(\frac{\Delta W}{\Delta P} \right) \times \frac{P}{W} = 10210.38604 \times \frac{6.2166}{7149.2771} = 8.878413893$$

Besides, the ΔD is -230.0643 as the price changed by $\pm .1$

$$W = \left(\frac{B}{P_W} \right) \left(\frac{a}{a + b} \right)$$

$$\frac{\partial D}{\partial P} = -\frac{B}{P^2} \cdot \frac{a}{a + b} = 1150.02366$$

$$\text{Then, the price elasticity of demand: } \varepsilon_d = -\varepsilon_d^M = \left(\frac{\Delta D}{\Delta P} \right) \times \frac{P}{D} = -1150.02366 \times \frac{6.2166}{7149.2771} = -1.0000$$

From a consumer standpoint, this good is classified as a unit elastic good. This means that for every 1% change in price, the quantity demanded changes by 1% in the opposite direction. Consumers' total expenditure remains constant regardless of price changes. The good is neither a luxury nor a necessity, but rather sits at the threshold between elastic and inelastic demand. Besides, the negative sign indicates some kind of income effect. Because a price rise is like a budget decrease, and a price lowering is like a budget increase.

Q1			Quantity				Quantity
Supply	Price(0)	6.216635	7149.277	Demand	Price(0)	6.216635	7149.277
	Price(+0.1)	6.316635	8237.861		Price(+0.1)	6.316635	7036.095
	Price(-0.1)	6.116635	6190.93		Price(-0.1)	6.116635	7266.16
		0.2	2046.931			0.2	-230.064
	Elasticity		8.878414		Elasticity		-1.0000

- 2) What is the income elasticity in the current equilibrium? What kind of good is this?

Ans.

$$W = \left(\frac{B}{P_W} \right) \left(\frac{a}{a + b} \right)$$

$$\frac{\partial W}{\partial B} = \left(\frac{1}{P_W} \right) \left(\frac{a}{a + b} \right) = 0.071492771$$

$$\text{The income elasticity: } \varepsilon_I = \left(\frac{\Delta Q}{\Delta I} \right) \times \frac{I}{Q} = 0.071492771 \times \frac{7149.277074}{100000} = 1.0000$$

This is a normal good with unitary income elasticity. A 1% increase in income leads to exactly a 1% increase in quantity demanded. The good's demand increases proportionally with income. It is neither a luxury good (which would have $\epsilon_I > 1$) nor a necessity (which would have $0 < \epsilon_I < 1$).

Q2		Quantity
Income(0)	1000	7149.277
Income(+1)	1001	7156.426
Income(-1)	999	7142.128
	2	14.29855
Elasticity		1.00000

3) What is the total profit to the industry in this situation?

Ans.

$$\text{Revenue} = \text{Price} \times \text{Quantity} = 6.2166 \cdot 7149.277 = 44444.4443$$

$$\text{Cost} = 38540.967 + 1377.976 = 39918.94247$$

$$\text{Total Profit} = \text{Revenue} - \text{cost} = 44444.4443 - 39918.94247 = 4525.501835$$

Q3			
Price	6.2166348	W1	6888.501
Quantily	7149.2771	W2	260.7758
Revenue	44444.444		
Cost 1	38540.967		
Cost 2	1377.9759		
Costs	39918.942		
Total Profit	4525.5018		

4) What is the Profit of each firm?

Ans.

For Firm 1:

$$\text{Profit} = \text{Revenue} - \text{Cost} = 42823.29621 - 38540.96659 = 4282.329621$$

For Firm 2:

$$\text{Profit} = \text{Revenue} - \text{Cost} = 1621.148088 - 1377.975875 = 243.1722132$$

Q4			
Price	6.2166348	W1	6888.501
Revenue 1	42823.296	W2	260.7758
Revenue 2	1621.1481		
Cost 1	38540.967		
Cost 2	1377.9759		
Profit 1	4282.3296		
Profit 2	243.17221		

5) SCENARIO A...make a copy of your supply-demand sheet and look at the following scenario:

The Demand function shifts because 30 more people come into the market (same utility). How much does price and quantity change? In terms of total demand does this make sense with respect to the elasticities?

Ans.

Now, $N' = N + 30 = 130$

After running the solver, new equilibrium price is 6.3839 with the quantity of 9050.5878

This change makes perfect sense given the elasticities: (1) With unitary income elasticity ($\epsilon_I = 1.00$), a 30% increase in market size (from 100 to 130 people) leads to a proportional increase in total market demand; (2) The price elasticity of demand being unitary ($\epsilon_d = -1.00$) explains why the price increase is relatively modest compared to the quantity increase; (3) The high price elasticity of supply ($\epsilon_s = 8.8995$) allows suppliers to respond effectively to the increased demand, leading to a significant quantity expansion with only a small price increase.

Q5		
N(0)	100	
N(+30)	130	
		Quantity
Price(0)	6.2166348	7149.277
Price(new)	6.3838702	9050.588
	-0.167235	-1901.31

- 6) SCENARIO B...make a copy of the Supply-Demand sheet...now, a third firm comes in with a production function given by $\alpha=.45$ $\beta=.4$. What happens to the equilibrium price, the total supplied, and the profit of each firm?

Ans.

With new firm added, the equilibrium price is now 6.1998 with total supply of 7168.7204,

For Firm 1:

Profit = Revenue – Cost = 41675.89157 – 37508.30241 = 4167.589157

For Firm 2:

Profit = Revenue – Cost = 1592.059249 – 1353.250362 = 238.8088873

For Firm 3:

Profit = Revenue – Cost = 1176.492736 – 1000.018825 = 176.4739103

Q6				
Firm 3:	alpha 3	beta 3		
	0.45	0.4		
			Quantity	
Price(0)	6.2166348			7149.277
Price(new)	6.1997736			7168.72
	0.0168611			-19.4433
Price	6.1997736	W1		6722.163
Revenue 1	41675.892	W2		256.7931
Revenue 2	1592.0592	W3		189.7638
Revenue 3	1176.4927			
Cost 1	37508.302			
Cost 2	1353.2504			
Cost 3	1000.0188			
Profit 1	4167.5892			
Profit 2	238.80889			
Profit 3	176.47391			
	4582.872			

- 7) If firms keep entering the market what would be the long run equilibrium profit and can you tell which of these three types of firms would dominate?

Ans.

In long run,

- (1) Perfect Competition Market: (a) Long-run economic profits are zero due to free entry and exit eliminating excess profits; (b) Price equals minimum average cost;
- (2) Monopolistic Competition Market: (a) Long-run economic profits are also zero due to no entry barriers; (b) However, price remains above minimum average cost;
- (3) Monopoly Market: (a) Can maintain economic profits in the long run; (b) Due to entry barriers, new firms cannot enter to share profits

Among these three market structures, monopoly market will dominate. Because it can continuously earn economic profits, and its effective entry barriers prevents new competitors, thus maintains pricing power.